Dataset	**Metric**	StdPT_StdFT	StdPT_PauseFT		PausePT_PauseFT	
			Prepending	Appending	Prepending	Appending
GSM8k	Acc.	7.5 ± 0.5	8.0 ± 1.0	6.9 ± 1.0	8.0 ± 0.4	8.5 ± 0.9
SQuAD	EM	36.4 ± 2.5	35.0 ± 1.5	40.2 ± 3.2	44.0 ± 3.2	55.9 ± 1.0
CommonQA	EM	26.9 ± 2.9	31.0 ± 1.3	28.8 ± 1.5	34.5 ± 1.0	34.8 ± 1.2
Lambada	EM	16.4 ± 1.7	17.8 ± 0.4	18.5 ± 0.6	18.0 ± 1.1	18.8 ± 0.1
PhysicalIQA	F1	73.3 ± 0.2	74.0 ± 0.3	74.0 ± 0.3	74.2 ± 0.2	74.2 ± 0.2
NaturalQ	EM	23.6 ± 1.2	24.1 ± 0.6	24.3 ± 1.4	25.7 ± 0.9	26.9 ± 0.4

Table 2: **Prepending vs. appending the pause tokens** (§5). We observe that prepending the pause tokens still outperforms the standard training baseline of StdPT_StdFT, but is suboptimal to appending the <pause> tokens for PausePT_PauseFT training algorithm. However, for StdPT_PauseFT, both have equal number wins and losses.

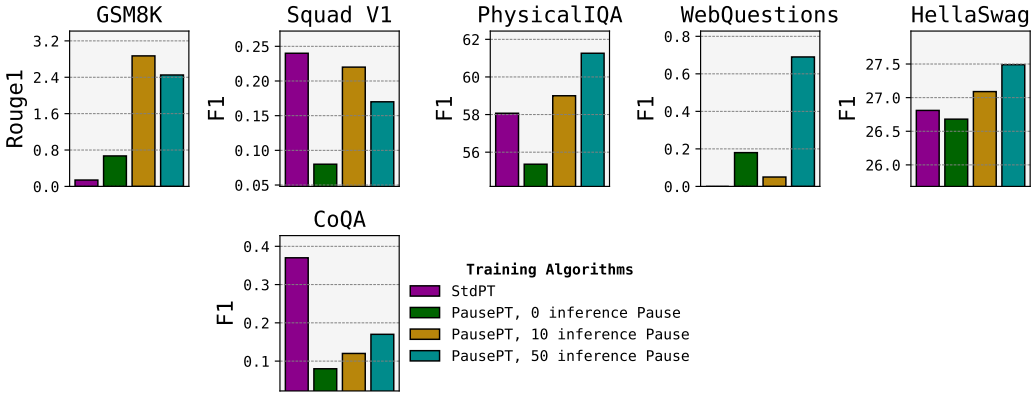


Figure 6: **Zero-shot evaluation of pause-pretrained models.** Zero-shot inference with pause tokens on a pause-pretrained model gives gains on tasks like GSM8k and HellaSwag. However, we note that our zero-shot accuracies are quite low, as we experiment with a small 1B parameter model.

E VARYING NUMBER OF PAUSE TOKENS M_{ft}

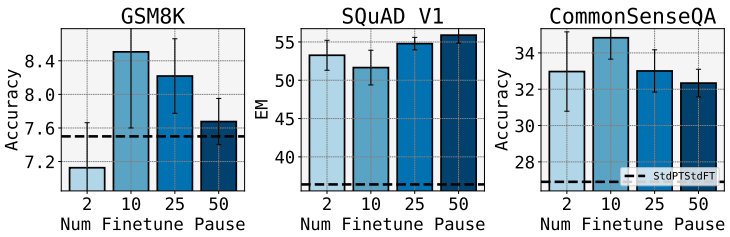
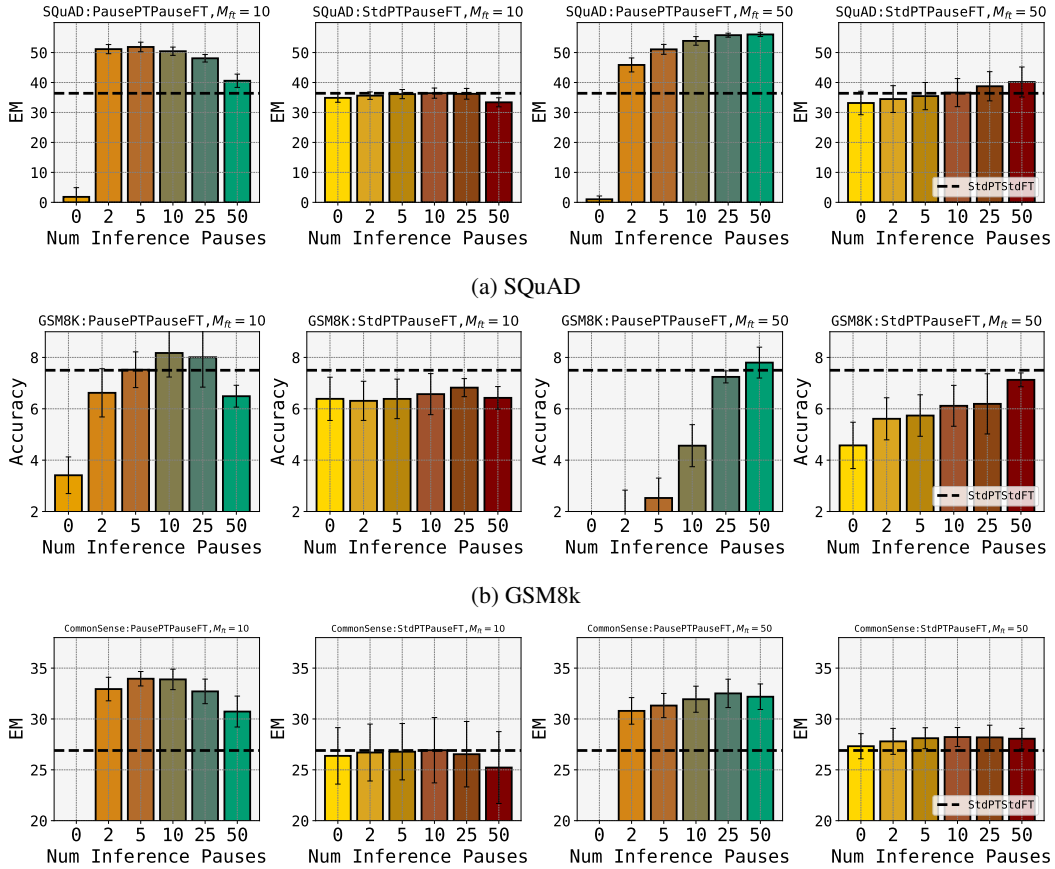


Figure 7: **Varying finetuning delay**: We examine the effect of varying the number <pause> tokens used in downstream finetuning (M_{ft}, §5) on the performance. Typically, we observe that there exists an optimal number of <pause> tokens as expected for each dataset.



(a) SQuAD

(b) GSM8k

(c) CommonSenseQA

Figure 8: **Varying inference-time delays:** We test the robustness of pause-trained models to varying number of inference time <pause> tokens (setting M_{inf} not equal to M_{ft}), which exposes the model to a serious test-time distribution shift (§5). Pause-training degrades gracefully to shifts as wide as $M_{\text{inf}} \in [5, 25]$ for $M_{\text{ft}} = 10$ and $M_{\text{ft}} = 50$ both for PausePT_PauseFT and StdPT_PauseFT, apart from GSM8k wherein there is a drop for $M_{\text{ft}} = 50$. In each row, the first and the third column considers the PausePT_PauseFT model for M_{ft} set to 10 and 50, respectively. Likewise, the second and the fourth column show the same for StdPT_PauseFT model.

In Figure 7, we study the effect of varying the number of pause tokens used during downstream finetuning (M_{ft}) on the downstream performance. We refer the reader to §5 for further details. Again we observe that there exists an optimal number of pause tokens to be used during downstream finetuning, depending on the task.

F Robustness to varying number of inference time pauses

Recall in §5 and Figure 4c we observed that pause-training is robust to using a different number of inference time pauses compared to that used during finetuning (i.e. $M_{\text{inf}} \neq M_{\text{ft}}$). We present additional results regarding the same in Figure 8a, Figure 8b and Figure 8c. Again, we observe that the performance degrades gracefully for the pause-trained models, even with shifts that halve the number of tokens seen. However, we still find a drastic drop in performance when no delay is given during inference for the PausePT_PauseFT model.

Dataset	**Learning Rate**	**Warmup Steps**	**Finetuning Steps**	**Batch Size**
SQuAD	1.0E-04	100	10000	256
GSM8k	1.0E-04	200	20000	16
HellaSwag	5.0E-06	100	1000	16
PhysicalIQA	1.0E-06	50	600	32
CoQA	5.0E-05	75	3500	16
CommonSenseQA	5.0E-05	100	4000	16
LAMBADA	5.0E-05	40	2800	16
WebQuestions	5.0E-04	200	2000	16
NaturalQuestions	1.0E-04	100	5000	256

Table 3: Downstream finetuning hyperparameters for the 1B model.

G Downstream Dataset Description

We finetune and evaluate the pretrained models (both standard and pause-pretrained) on the following datasets:

1. GSM8k: A reasoning task with 8.5k grade school math word problems (Cobbe et al., 2021).
2. SQuAD V1: Reading-comprehension task based on Wikipedia (Rajpurkar et al., 2016).
3. CommonSenseQA: Requires different types of commonsense knowledge to choose the correct answer (Talmor et al., 2019). Our implementation of CommonSenseQA is as a decoding task, and hence we report Exact Match (EM) scores.
4. LAMBADA: Text-understanding task requiring last-word prediction based on a long context (Paperno et al., 2016).
5. Web Questions: A fact-recall dataset of commonly-asked questions on the web (Berant et al., 2013).
6. PhysicalIQA: A physical commonsense reasoning dataset, which test the ability to understand interactions with the world (Bisk et al., 2020).
7. Natural Questions: QA task which requires answering fact-based questions from Wikipedia article pages (Kwiatkowski et al., 2019). Since we use the closed-book version of this dataset (no access to helpful context), this is a fact-recall task.
8. HellaSwag: Next-sentence prediction task based on common-sense inference (Zellers et al., 2019).
9. CoQA: Question-answering task based on a context (Reddy et al., 2019).

H Hyperparameters: Downstream finetuning

We share all the hyperparameters for downstream finetuning in Table 3 (1B model) and Table 4 (130M model). We also provide the decoder-only architecture details for the two models considered in this work in Table 5.